# Income-Stratified PCE Price Indices (ISPCEPX)

Distribution of Personal Income Statistics

## Overview

The ISPCEPX (Income-Stratified PCE Price Indices) table reports Fisher chain-linked price indices constructed separately for each decile and quintile of the equivalized disposable personal income (DPI) distribution. Each index uses the expenditure composition of that income group as weights rather than economy-wide expenditure shares, so cross-group differences in the indices reflect differences in the composition of consumption expenditures at different points in the distribution. The methodology follows Gindelsky and Martin (forthcoming). The table covers calendar years 2000–2024. All indices except the published PCE PI are set equal to 100 in 2000.

A central feature of this approach is that the price indices are based on Personal Consumption Expenditures (PCE) from the national accounts rather than the Consumer Price Index (CPI) market basket. PCE is broader than the CPI market basket, which mainly includes out-of-pocket household expenditures. PCE also includes expenditures made on behalf of households, such as government and employer contributions to health insurance premiums, as well as imputed financial services (FISIM – financial intermediation services indirectly measured). These components are absent from CPI-based distributional price indices. Because imputed financial services have experienced high inflation and are concentrated at the top of the income distribution, their inclusion compresses measured inflation inequality relative to CPI-based estimates.

The ISPCEPX are directly tied to the joint distribution of DPI and PCE published on this website. The expenditure shares used to construct the ISPCEPX are derived from the same CPS microdata and NIPA-scaled PCE aggregates that underlie the joint distribution statistics, and the income grouping is identical. Accordingly, the ISPCEPX can be used to convert the nominal PCE shares in the joint distribution tables to real consumption expenditures shares, and to assess how differences in price levels across the distribution affect measured consumption expenditures inequality.

## Column Descriptions

1. **Year**

Calendar year (income year).

2. **Published PCE PI**

The aggregate PCE price index published by BEA in NIPA Table 2.4.4U, line 1. Expressed in the BEA standard base (2017 = 100).

3. **Indexed Published PCE PI**

The same published aggregate PCE price index reindexed to 2000 = 100, to facilitate direct comparison with the ISPCEPX columns.

4. **ISPCEPX: 0-10% through ISPCEPX: 90-100%**

Fisher chain-linked price indices for each decile of the equivalized DPI distribution, indexed to 2000 = 100. Deciles are defined by ranking households on equivalized DPI (household DPI divided by the

square root of household size), using survey weights. ISPCEPX: 0-10% corresponds to the lowest-income decile and ISPCEPX: 90-100% to the highest.

**5. ISPCEPX: 0-20%, 20-40%, 40-60%, 60-80%, 80-100%**

Fisher chain-linked price indices for each quintile of the equivalized DPI distribution, indexed to 2000 = 100. These are computed directly from the combined expenditure shares of the two constituent deciles, not as averages of the decile indices.

## Methodology

The following summarizes the construction of the ISPCEPX; see Gindelsky and Martin (forthcoming) for full details. Group-specific expenditure shares across are constructed for approximately 150 detailed PCE categories and scaled to match NIPA PCE category totals from Table 2.4.5U, consistent with the joint distribution statistics published on this website. These shares are combined with the corresponding NIPA chain-type price indices from Table 2.4.4U.

The Fisher price index for each group is chain-linked annually. The year-over-year link for group g in year t is the geometric mean of the Laspeyres index (using year t–1 expenditure shares as weights) and the Paasche index (using year t shares), both evaluated at the detailed PCE category level. Since all groups face the same category-level price indices from the national accounts, cross-group differences in the ISPCEPX reflect entirely differences in expenditure composition across the distribution. The chain-linked index is accumulated from the base year (2000 = 100).

At the time of publication, the joint income-expenditure distributions were not yet available for 2024. To impute 2024 expenditures by group, the following procedure was used: for each of 150 detailed PCE categories, the income group's share of total category spending from 2023 was applied to the 2024 NIPA category total. For instance, the bottom decile of income had about 3% of the PCE total for new domestic autos in 2023. Accordingly, the bottom decile's 2024 spending on new domestic autos was imputed as 3% of the PCE total for new domestic autos in 2024.

The income grouping variable is equivalized DPI, consistent with the ranking used throughout the joint distribution. Decile boundaries are determined annually using household survey weights.

## Interpretation

Divergence across ISPCEPX columns implies that the price deflator relevant for a given income group differs from both the aggregate PCE deflator or from deflators relevant for other groups. The magnitude of cross-group divergence depends on the degree of expenditure heterogeneity across the distribution and on the extent to which relative price changes are concentrated in categories that are over- or under-represented in particular income groups' budgets. Using a common (aggregate) deflator to compute real consumption expenditures across groups will therefore misstate changes in real consumption expenditures inequality to the extent that group-specific price indices diverge from the aggregate. When the ISPCEPX for a lower-income group rises faster than that for a higher-income group over a given period, households at the bottom of the distribution face a higher effective inflation rate.

## Reference

Gindelsky, Marina and Robert Martin. "Rethinking Inflation Inequality: Evidence from National Accounts." Forthcoming, *Macroeconomic Dynamics*. Preliminary versions: BEA Working Paper 2025-9; BLS Working Paper WP-586.